Introduction to psycholinguistics

Day 1: Language, linguistics, psycholinguistics

Deniz Özyıldız

Stuttgart
Oct. 19th, 2026

Small little experiment #1

The McGurk effect

Do you remember what the main articulatory differences are between the sounds /b/, /v/, /d/ and /g/?

The McGurk effect

Do you remember what the main articulatory differences are between the sounds /b/, /v/, /d/ and /g/?

Do you think that visual cues are important in how we perceive speech sounds?

The McGurk effect

Do you remember what the main articulatory differences are between the sounds /b/, /v/, /d/ and /g/?

Do you think that visual cues are important in how we perceive speech sounds?

With this in mind, let's take a look at the *McGurk effect*.

The McGurk effect

Click on one of the first two links and listen to the audio with your eyes closed. Then re-watch the video with your eyes open.

female voice without commentary
`https://youtu.be/ZZERYAg2Gkk`

male voice without commentary
`https://youtu.be/PWGeUztTkRA`

some commentary
`https://youtu.be/2k8fHR9jKVM`

Describe in your own words what seems to be going on.
What else would you be interested in testing?

The McGurk effect

Hearing lips and seeing voices

MOST verbal communication occurs in contexts where the listener can see the speaker as well as hear him. However, speech perception is normally regarded as a purely auditory process. The study reported here demonstrates a previously unrecognised influence of vision upon speech perception. It stems from an observation that, on being shown a film of a young woman's talking head, in which repeated utterances of the syllable [ba] had been dubbed on to lip movements for [ga], normal adults reported hearing [da]. With the reverse dubbing process, a majority reported hearing [bagba] or [gaba]. When these subjects listened to the soundtrack from the film, without visual input, or when they watched untreated film, they reported the syllables accurately as repetitions of [ba] or [ga]. Subsequent replications confirm the reliability of these findings; they have important implications for the understanding of speech perception.

McGurk & MacDonald (1976)

Small little experiment #2

In the next couple of slides, you're going to see some sentences presented word by word.

Your task is to say whether they are acceptable sentences of English.

Practice item 1

+

Practice item 1

The

Practice item 1

road

Practice item 1

to

Practice item 1

the

Practice item 1

city

Practice item 1

is

Practice item 1

full

Practice item 1

of

Practice item 1

enemies.

Practice item 2

+

Practice item 2

The

Practice item 2

silly

Practice item 2

to

Practice item 2

the

Practice item 2

water

Practice item 2

are

Practice item 2

devoid

Practice item 2

of

Practice item 2

people.

The

key

to

the

cabinets

are

on

the

table.

The

keys

to

the

cabinet

is

on

the

table.

The

keys

to

the

cabinets

are

on

the

table.

The sentences

Practice

(1) a. The road to the city is full of enemies.
b. The silly to the town are devoid of people.

‘Agreement attraction’

(2) a. The key to the cabinets are on the table.
b. The keys to the cabinet is on the table.

Control

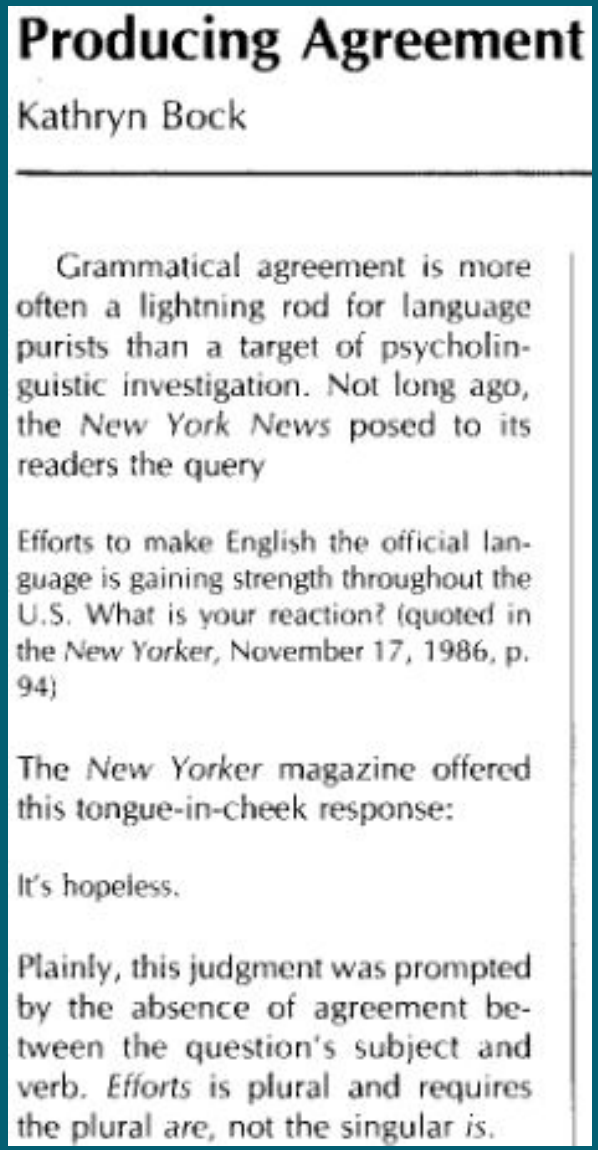
(3) The keys to the cabinets are on the table.

What do you think is going on?

Do you think it makes a difference whether the words are presented one by one vs. the sentences as a whole?

What other sentences would you like to look at?

Agreement attraction



Producing Agreement

Kathryn Bock

Grammatical agreement is more often a lightning rod for language purists than a target of psycholinguistic investigation. Not long ago, the *New York News* posed to its readers the query

Efforts to make English the official language is gaining strength throughout the U.S. What is your reaction? (quoted in the *New Yorker*, November 17, 1986, p. 94)

The *New Yorker* magazine offered this tongue-in-cheek response:

It's hopeless.

Plainly, this judgment was prompted by the absence of agreement between the question's subject and verb. *Efforts* is plural and requires the plural *are*, not the singular *is*.

Bock (1995)

What is language?